

AUDIO SLIDERS: MEASURING WHAT A SLIDER DOES TO MUSIC

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ABSTRACT

A slider is a low-rank update to a frozen generative model whose strength is set at inference time, so one prompt and seed can be moved along a single axis. Audio sliders are scored by text–audio embedding similarity. We ask three questions that leaves open: does a property measured from the waveform follow the slider, is the result still the same piece, and does it still sound like music. We train sliders on Stable Audio Open 1.0 and ACE-Step 1.5 in two ways: from a prompt pair, and from two sets of the model’s own clips with no text, sorted by a descriptor, an aesthetics score, or a principal direction of their embeddings. On 24 held-out prompts, sliders for mood, harmony, ensemble size, and melody move their descriptors monotonically while content enjoyment stays above the mean of 2,000 real recordings. Switching a slider on after the first 21 of 50 sampling steps raises similarity to the unsteered piece from 0.67 to 0.85. Sliders for note density and percussion on Stable Audio Open pass the embedding test and fail the waveform test. Principal components of one concept’s clips carry up to 25% of their variance and read as musical contrasts.

1. INTRODUCTION

A text-to-music model takes a prompt and returns a clip. A musician who likes the clip but wants it a little happier, or fuller, or with richer harmony, has to rewrite the prompt and accept a different piece. A slider offers something else: keep the prompt and the seed, and move the result along one axis by a chosen amount.

Sliders exist for image diffusion as low-rank adapters with a runtime scale [1], and the idea of discovering the axes themselves, with no names supplied, comes from SliderSpace [2]. Both have reached audio models as baselines [3, 4]. In that work a slider counts as working when the output moves toward the attribute text in a joint text–audio embedding and stays perceptually close to the original.

For music that leaves three things unmeasured. The embedding can move without the audio changing in the way a listener would name: Section 5 shows two sliders whose embedding score rises while the descriptor they are named

after stays flat. A slider can reach its target by wrecking the clip, and similarity to the original does not say whether what remains is music. And nothing in the score says what else moved.

This paper is about measuring those things, and about a second way to train a slider that follows from the measurements. Its contributions:

- A protocol that scores a slider on a waveform descriptor fixed in advance, on a reference-free quality predictor anchored by real recordings, on similarity to the unsteered piece, and on a tag panel that puts the change in words. It reports effect only over the *usable span*, the positions at which quality has not yet dropped.
- Evidence on two open backbones, a diffusion model and a few-step rectified-flow model, that prompt-pair sliders for musical attributes pass this protocol, and that two acoustic ones pass the embedding test and fail the waveform test.
- A trainer that needs no text: one low-rank update trained with opposite signs on two sets of the model’s own clips. It turns any per-clip measurement into a slider, including a quality score and discovered principal directions, and does not need a differentiable path through the decoder and the embedding model.
- A measurement of timestep gating: leaving the earliest sampling steps unsteered keeps the piece and its quality at a known cost in range.

2. RELATED WORK

Sliders for image diffusion. Concept Sliders [1] train a LoRA [5] whose output matches the frozen model’s prediction shifted along the difference between a positive-prompt and a negative-prompt prediction. SliderSpace [2] removes the prompt pair: it takes principal components of CLIP embeddings of many images generated for one concept and trains one rank-one LoRA per component, using a few-step distilled model so the embedding loss can see a clean image after a single denoising step.

Sliders and steering for audio. FreeSliders [3] apply the Concept Sliders target at inference time without training, on image, video, and audio models including Stable Audio Open; most of their audio concepts are sound effects.



TADA [4] steers activations in audio diffusion transformers and reports that concepts concentrate in a few attention layers; it includes Concept Sliders as a baseline. Facchiano et al. [6] compute difference-in-means steering vectors for tempo and timbre in a music model. AnchorSteer [7] trains text-free concept modules by reconstruction on self-generated clips, the nearest relative of our set trainer. All of these score steering by text–audio alignment in a learned embedding and by perceptual distance to the unsteered output; TADA additionally checks two concepts against a descriptor, and Wang [8] shows with a key estimator and a beat tracker that apparent agreement with an instruction can be the model’s prior. None scores every control against a waveform property fixed in advance, none compares against plain signal processing, and none reports what else moves when a control moves.

Measured controls and discovered features. Measured Sliders [9] define image sliders by differentiable measurements and train them to move the target while holding other measurements still; we adopt their selectivity score. Our audio descriptors are not differentiable through the decoder, so we sort clips by the measurement and train between the two ends. Guinot et al. [10] learn sparse features of music embeddings to steer retrieval; no one has trained generator sliders from axes of real-music embeddings.

Effects with known parameters. RIME [11] builds paired edit data for music post-production by applying effects with known parameters. We borrow the idea in a different role: effects with known parameters are a baseline that a learned control has to justify itself against.

3. METHOD

Backbones. We use two frozen text-to-music models behind one interface. Stable Audio Open 1.0 [12] is a latent diffusion transformer with a v -prediction objective: $z_t = \alpha_t x_0 + \sigma_t \epsilon$ with $\alpha_t = \cos(\pi t/2)$, $\sigma_t = \sin(\pi t/2)$, and the network predicts $v = \alpha_t \epsilon - \sigma_t x_0$. We sample with DPM-Solver++(2M) for 50 steps and classifier-free guidance 7. ACE-Step 1.5 XL turbo [13] is a rectified-flow transformer, $x_t = (1 - t)x_0 + t\epsilon$ with the network predicting $\epsilon - x_0$, distilled to 8 Euler steps without guidance. Everything below is written for a generic prediction $f_\theta(z_t, t, c)$ and applies to both.

A slider. A slider is a set of low-rank updates, one per linear layer in the self-attention, cross-attention, and feed-forward blocks of the transformer. With slider position $s \in \mathbb{R}$ a layer computes

$$y = Wx + s \frac{\alpha}{r} BAx, \quad (1)$$

with rank $r = 4$ and $\alpha = 1$. s is read at every forward pass and can differ per sample, so one batch can hold one clip at several positions. Several sliders can be attached at once and their updates add.

Training from a prompt pair. Given a base prompt c and two phrases describing the ends of an attribute, let c^+ and

c^- be c with each phrase appended. Following Concept Sliders we train the slider so that, for $s \in \{+1, -1\}$,

$$f_{\theta,s}(z_t, t, c) \approx f_\theta(z_t, t, c) + s \eta (f_\theta(z_t, t, c^+) - f_\theta(z_t, t, c^-)), \quad (2)$$

with all right-hand terms from the frozen model and a squared-error loss. Both signs are trained in every batch. z_t is obtained by sampling from noise for a random number of steps with the slider active, so the slider is trained on states it will visit. We use $\eta = 4$ for Stable Audio Open and $\eta = 2$ for ACE-Step, 1,000 iterations, batch 4, and 48 training prompts that share no prompt with the evaluation set.

Training from two sets of clips. The second trainer uses no text. Let H and L be two sets of latents. We minimise the ordinary denoising loss of the model with the slider at $s = +1$ on H and at $s = -1$ on L :

$$\mathbb{E}_{x_0 \in H} \|f_{\theta,+1}(z_t, t, c) - u\|^2 + \mathbb{E}_{x_0 \in L} \|f_{\theta,-1}(z_t, t, c) - u\|^2, \quad (3)$$

where u is the model’s regression target for (x_0, ϵ, t) . One update serves both ends with opposite sign. To first order, anything the two sets share pulls the update in opposite directions and cancels, and what remains is the difference between the conditional mean predictions of the two sets, which is the direction classifier guidance toward H and away from L would take. We build H and L from the model’s own output: generate a corpus of clips, measure something on each clip, and take the top and bottom 30% *within each prompt* so that prompts are balanced across the two sets. The measured quantity can be a signal descriptor, an aesthetics score, or the projection on a principal direction of CLAP embeddings. The last choice is unsupervised discovery in the sense of SliderSpace, without requiring a differentiable path through the decoder and the embedding model.

Timestep gating. At inference the slider can be switched on only for $t \leq t_0$. The earliest steps decide the layout of the piece; leaving them unsteered keeps it. Section 5 measures the effect.

4. MEASUREMENT

For each prompt and seed we render the same initial noise at every slider position and call the result a trajectory. All reported numbers use 24 evaluation prompts held out from training.

Does a measured property follow the slider? Each slider is paired in advance with a descriptor computed from the waveform with no learned model: spectral centroid, onset rate, pulse clarity, energy decay time, percussive energy share, low-band and stereo-side energy ratios, key clarity, major-minus-minor key fit, harmonic change rate, and level dynamics. We report the mean Spearman correlation between position and descriptor over trajectories (ρ), the share of trajectories whose two ends are ordered

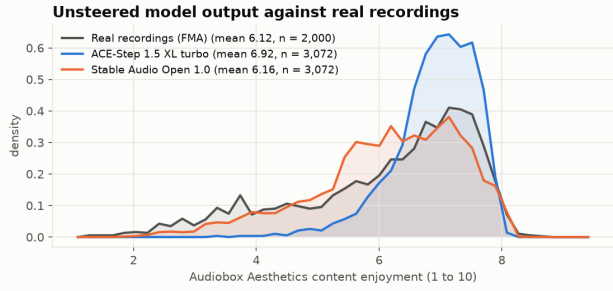


Figure 1. Content enjoyment of unsteered model output and of real recordings.

as intended, and the change in the descriptor in units of its standard deviation across unsteered clips.

Does it still sound like music? Every clip is scored by the Audiobox Aesthetics predictor [14]. We use content enjoyment (CE) on its 1–10 scale and anchor it with 2,000 ten-second excerpts of real recordings from FMA [15]. The *usable span* of a slider is the set of positions, walking outward from zero, over which mean CE stays within 0.5 of the unsteered clips. Effect sizes are reported inside the usable span, which stops a slider from being credited for change it only achieves by destroying the clip.

What does it change, in words? We embed every clip with CLAP [16] and score it against 80 tags for instruments, genres, moods, and playing characteristics. For a slider we report the tags whose similarity rises and falls most between its two ends, and the rank correlation between position and the projection on the slider’s text direction. The same correlation is computed with MuQ-MuLan as a check that does not depend on CLAP.

How much of the piece survives? CLAP similarity, frame-wise chroma similarity, and onset-envelope correlation between each clip and the same trajectory at position zero.

What else moves? The slope of every descriptor against position, in standard deviations per unit, for every slider: a leakage matrix.

Baselines. Under identical prompts, seeds, and sampler: the training target applied directly at every step (the training-free approach of FreeSliders, at three times the compute), interpolation of the text conditioning toward c^+ or c^- , and, where one exists, a conventional effect applied to the unsteered clip (spectral tilt, low shelf, mid/side gain, convolution reverb, soft clipping, band-limiting).

5. RESULTS

The yardstick. Audiobox content enjoyment averages 6.12 over 2,000 real recordings. Unsteered ACE-Step clips average 6.92 and unsteered Stable Audio Open clips 6.16 (Figure 1). A slider that drops a clip well below the real-music mean has left the region in which the change it made is worth having.

Musical sliders on ACE-Step. Table 1 and Figure 2 report prompt-pair sliders on 24 held-out prompts, three seeds,

nine positions. Where a descriptor was assigned in advance it rises with the slider: harmonic change rate for harmony, production complexity for ensemble size, key clarity for melody, and major-minus-minor fit for mood. The tag panel agrees in words: the mood slider raises *happy* and lowers *dark-toned*; the groove slider raises *latin* and *funk* and lowers *choir* and *strings*; the ensemble slider raises *orchestral film score* and *choir*. Content enjoyment stays above the real-music mean across the range for all but the tension slider, which collapses past +1 and whose usable span ends there. The tags and the direction score both come from CLAP, so we repeat the direction test with MuQ-MuLan, which shares no weights or training data pipeline with it: every slider is monotone under it as well (Table 1).

Gating. On Stable Audio Open a prompt-pair slider at ± 1 moves its descriptor a long way and takes the piece with it. Table 2 and Figure 3 show what happens when the slider is switched on only after the first steps. With the slider on from step 21 the brightness slider keeps 1.41 of its 2.37 standard deviations of range, similarity to the unsteered clip rises from 0.67 to 0.85, and the loss in enjoyment falls from 1.0 to 0.3 points. The other three sliders tested follow the same curves.

The embedding says yes, the waveform says no. The same gating runs include two sliders trained from the prompt pairs for note density and for percussion. Both move the output along their CLAP text direction at every gate setting. Neither moves its descriptor: the rank correlation between position and onset rate or percussive energy share never exceeds 0.37, and the range is near zero. Scored by embedding alone, both would be reported as working. The descriptor has to be sound as well: with a tempo estimate prone to octave errors the tempo slider scored $\rho = -0.01$, and with a beat tracker it scores 0.57 and moves the mean tempo from 90 to 167 BPM.

Stable Audio Open, all sliders. Table 3 gives the gated evaluation of every prompt-pair slider.

Discovered axes. For each of five concepts we generated 1,024 ACE-Step clips and took principal components of their CLAP embeddings. The leading component carries between 16% and 25% of the variance within a concept (Figure 4), and the tags aligned with each end read as musical contrasts (Table 4).

From axes to sliders. For three concepts we trained the four leading components into sliders with the set trainer and tested each on 32 fresh seeds of its concept. All 12 follow the axis they were trained on, with rank correlations between 0.43 and 0.81, and the tags that move in the output match the labels each axis received from the corpus (Table 5). Unlike the prompt-pair sliders these stay usable only to about ± 1.5 .

6. LIMITS

No listening test was run; every claim about quality rests on a learned predictor and every claim about meaning on

Slider	Descriptor	ρ	Ordered	MuQ ρ	Span	Moved	Kept	CE ₀	CE _{ends}
brightness	centroid_oct	0.97	1.00	0.80	-1.0 to +2.0	4.41	0.74	6.95	6.04
density	onset_rate	0.78	0.92	0.86	-0.5 to +2.0	1.25	0.84	6.95	6.09
mood	majorness	0.39	0.75	0.88	-1.0 to +2.0	0.87	0.79	6.95	6.65
harmony	harmonic_change	0.41	0.76	0.73	-1.5 to +2.0	0.84	0.83	6.95	6.64
groove	—	—	—	0.56	-1.5 to +2.0	—	0.79	6.95	6.88
ensemble	pc	0.60	0.90	0.77	-1.5 to +2.0	0.92	0.80	6.95	6.56
tension	—	—	—	0.86	-1.5 to +1.0	—	0.83	6.95	5.64
melody	key_clarity	0.48	0.78	0.89	-1.0 to +2.0	0.97	0.80	6.95	6.47

Table 1. Prompt-pair sliders on ACE-Step 1.5 XL turbo. ρ : mean rank correlation between position and descriptor. Ordered: share of trajectories with correctly ordered ends. MuQ ρ : the same correlation for the MuQ-MuLan direction score. Span: usable span. Moved: descriptor change over the span in standard deviations of unsteered clips. Kept: CLAP similarity to the unsteered clip at the ends of the span. CE: content enjoyment unsteered and at the extreme positions.

On after	Moved	CLAP	Chroma	CE
0	2.37	0.67	0.68	5.15
10	2.11	0.71	0.71	5.34
14	1.98	0.74	0.74	5.45
21	1.41	0.85	0.84	5.87
29	0.42	0.97	0.97	6.08

Table 2. Timestep gating for the brightness slider on Stable Audio Open at positions ± 1 : descriptor moved (std), CLAP and chroma similarity to the unsteered clip, content enjoyment. Unsteered CE is 6.16.

CLAP tags and signal descriptors. Descriptors are proxies: energy decay time stands in for reverberation and is confounded by sustained instruments, and estimated tempo suffers octave errors. Clips are ten seconds long. The usable-span tolerance of 0.5 CE is a choice, and a different tolerance moves the reported spans. The set-contrast trainer’s cancellation argument is first-order; correlated properties of the two sets are learned together, which the leakage matrix shows. Discovered axes are labelled after the fact by the tags they move, and a different tag vocabulary would produce different labels for the same direction.

7. REFERENCES

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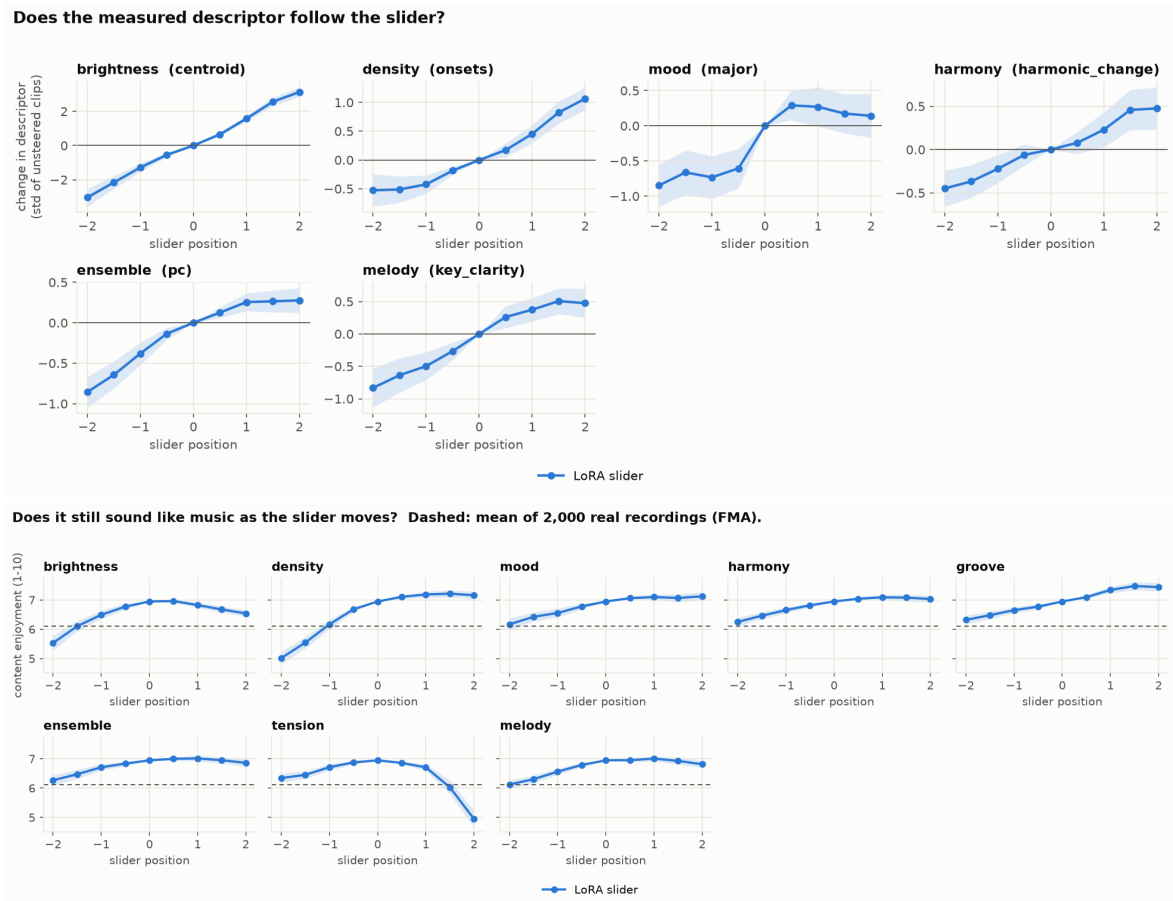


Figure 2. ACE-Step. Top: change in each slider’s descriptor against position, in standard deviations of unsteered clips. Bottom: content enjoyment against position; the dashed line is the mean of real recordings.

[16] Y. Wu, K. Chen, T. Zhang, Y. Hui, T. Berg-
Kirkpatrick, and S. Dubnov, “Large-scale contrastive
language-audio pretraining with feature fusion and
keyword-to-caption augmentation,” in *ICASSP*, 2023,
arXiv:2211.06687.

Turning the slider on later keeps the piece and its quality, and costs range

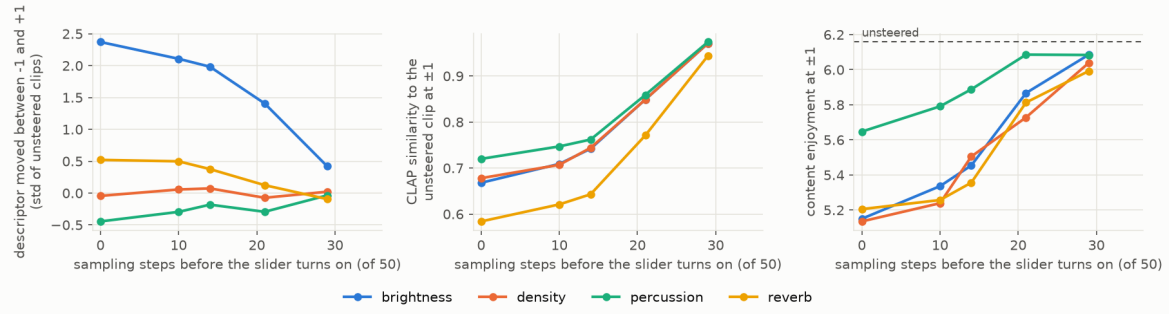


Figure 3. Timestep gating on Stable Audio Open for four sliders.

Slider	Descriptor	ρ	Ordered	MuQ ρ	Span	Moved	Kept	CE ₀	CE _{ends}
brightness	centroid_oct	0.84	0.96	0.53	-1.0 to +1.5	1.74	0.82	6.04	5.42
density	onset_rate	0.30	0.75	0.60	-1.5 to +1.0	-0.00	0.82	6.04	5.50
reverb	decay_s	0.37	0.79	—	-1.0 to +1.0	0.66	0.77	6.04	5.15
energy	flux	0.71	0.94	0.72	-1.5 to +0.5	0.72	0.82	6.04	5.15
tempo	beat_bpm	0.57	0.78	—	-1.5 to +1.0	1.21	0.80	6.04	5.47
percussion	percussive_ratio	0.34	0.71	—	-1.5 to +1.5	-0.04	0.77	6.04	5.92
distortion	flatness	0.50	0.81	0.17	-1.5 to +0.5	0.30	0.89	6.04	5.44
lofi	rolloff_oct	0.28	0.68	—	-1.5 to +0.5	0.28	0.85	6.04	4.91
bass	bass_ratio	0.92	0.97	0.59	-1.0 to +1.0	1.68	0.85	6.04	5.37
width	side_ratio	0.95	1.00	—	-1.0 to +1.5	4.70	0.80	6.04	5.53
mood	majorness	-0.04	0.43	—	-1.5 to +1.5	-0.28	0.79	6.04	5.85
electronic	—	—	—	0.52	-1.5 to +1.5	—	0.74	6.04	6.02
harmony	harmonic_change	0.19	0.64	—	-1.0 to +1.5	0.06	0.83	6.04	5.54
groove	—	—	—	0.07	-0.5 to +1.5	—	0.85	6.04	5.79
articulation	decay_s	0.33	0.74	0.22	-1.0 to +1.0	0.43	0.87	6.04	5.48
ensemble	pc	0.70	0.93	0.35	-1.0 to +1.5	0.94	0.81	6.04	5.66
tension	—	—	—	—	-1.5 to +1.0	—	0.84	6.04	5.65
vintage	—	—	—	—	-1.5 to +1.0	—	0.82	6.04	5.50
melody	key_clarity	0.26	0.65	—	-1.0 to +1.5	0.38	0.83	6.04	5.84
dynamics	dynamics_db	0.26	0.67	-0.22	-1.0 to +1.0	0.57	0.86	6.04	5.44

Table 3. Prompt-pair sliders on Stable Audio Open 1.0 with the slider switched on after step 21 of 50. Columns as in Table 1.

Concept	Var.	Toward	Away
music	25.0%	latin, repetitive, happy	lo-fi, hip hop, electric piano
music	11.0%	organ, complex, improvised	lo-fi, calm, drums
piano music	25.3%	lo-fi, distorted, hip hop	acoustic guitar, reggae, happy
piano music	14.0%	blues, loud, playful	dreamy, ambient, reverberant
electronic dance music	19.9%	vocals, techno, bright	melancholic, classical, improvised
electronic dance music	12.1%	swung, hip hop, flute	romantic, complex, melodic
orchestral music	16.4%	acoustic guitar, romantic, bass guitar	choir, organ, distorted
orchestral music	10.2%	simple, clean, calm	orchestral film score, epic, cello
guitar music	17.8%	latin, repetitive, happy	mysterious, distorted, tense
guitar music	13.4%	strings, harp, acoustic	saxophone, funk, piano

Table 4. The two leading principal components of CLAP embeddings for each concept (1,024 ACE-Step clips each), labelled by the tags they align with.

A few axes carry much of how clips of one concept differ (ACE-Step, 1,024 clips each)

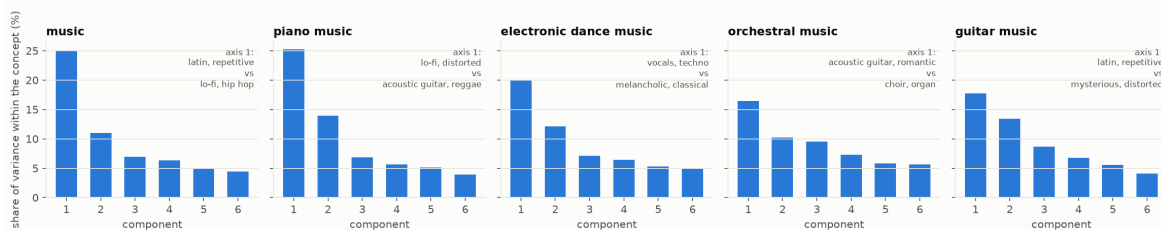


Figure 4. Share of within-concept variance carried by the first six components.

Concept	Corpus label (toward / away)	ρ	Kept	Tags that rise / fall in the slider's output
music	latin, repetitive / lo-fi, hip hop	0.76	0.74	latin, repetitive / trumpet, saxophone
music	organ, complex / lo-fi, calm	0.61	0.78	acoustic, acoustic guitar / trumpet, epic
piano music	lo-fi, distorted / acoustic guitar, reggae	0.81	0.79	distorted, dark-toned / latin, reggae
piano music	blues, loud / dreamy, ambient	0.72	0.73	hip hop, lo-fi / orchestral film score, cello
electronic dance music	vocals, techno / melancholic, classical	0.72	0.62	techno, pop / choir, organ
electronic dance music	swung, hip hop / romantic, complex	0.61	0.69	cello, vocals / funk, happy

Table 5. Sliders trained along discovered axes (two leading components per concept). ρ : rank correlation between slider position and the output's projection on the axis, over 32 fresh seeds. Kept: CLAP similarity to the unsteered clip at the ends of the usable span.